

SelfBlending: Artificial Intelligence-Driven Augmentation With Hand Interactions for Seamless Reality Blending in Virtual Environments

Ahmed Elsharkawy , Bocheon Gim , Aya Ataya , and SeungJun Kim , *Member, IEEE*

Abstract—Accessing real-world objects during immersive virtual reality (VR) experiences remains challenging, as current cross-reality systems often rely on predefined interaction steps, tracking devices/markers, or fixed object setups. They also lack support for personalized object recall, where users can add, remove, or modify real-world items blended into the virtual environment (VE). Many head-mounted displays (HMDs) include passthrough technology to switch between virtual and real worlds, but it often disrupts immersion by requiring a full shift from virtual to real. Thus, maintaining an optimal balance between virtuality and reality is difficult. To address these challenges, we developed SelfBlending, a framework that uses AI-based hand tracking to let users label physical objects through freehand gestures, then blends the selected item into the VE using object recognition, enabling interaction with the relevant real-world object. SelfBlending was evaluated against two common interaction conditions: the default passthrough feature in VR HMDs and the conventional approach of physically removing the HMD to access real-world objects. Results from seated, single-object interactions with tabletop-placed items showed that SelfBlending enhanced user experience by boosting presence, supporting efficient physical interaction, and improving cross-reality continuity. It also enabled selective interaction with real objects while minimizing the disruption of VR experience.

Index Terms—Cross-reality interaction, virtual reality, augmented virtuality, passthrough, physical object interaction.

I. INTRODUCTION

A PERSISTENT challenge in virtual reality (VR) is the inadequacy of transition techniques, which causes users to remain immersed in the virtual environment (VE) while interacting with real-world objects. Users lose real-world awareness when interacting in the VE, which makes simple tasks difficult and disruptive. Users have to disengage from VR to interact with the real world when performing tasks, breaking immersion; for instance, a gamer may need to drink water, a professional in VR training may need to take notes or access tools, or a designer in a VR architectural simulation might want to sketch on paper or a tablet. Previous studies [1], [2] have shown that even brief real-world interruptions can disrupt user presence and continuity while in VR.

In addition, existing techniques for integrating physical objects into VEs remain limited in flexibility and everyday usability. Prior work has demonstrated selective incorporation of physical objects through appearance-based cues such as shape or color [3], [4], which often requires prior tuning or visual augmentation of objects, or through marker-based and hardware-assisted tracking that can be reliable but typically assumes controlled setups and preconfigured objects [5], [6], [7]. Such approaches require attaching external markers or tracking hardware to each object a user wishes to access, which is impractical for many everyday objects and can interfere with natural handling, as grasping or occluding the attached markers may reduce tracking robustness and user comfort. Proximity-based blending further constrains interaction by requiring objects to be held within a specific interaction range to become available for blending [8], which limits how users can locate, retrieve, and access physical objects during an ongoing VR experience. More recent systems employ CNN-based vision models [9], [10]; however, many rely on dense, continuous semantic segmentation over the entire field of view and on predefined object classes. Although effective for general scene understanding, such designs introduce sustained computational overhead and may expose large portions of the physical environment even when the user intends to access only a single object.

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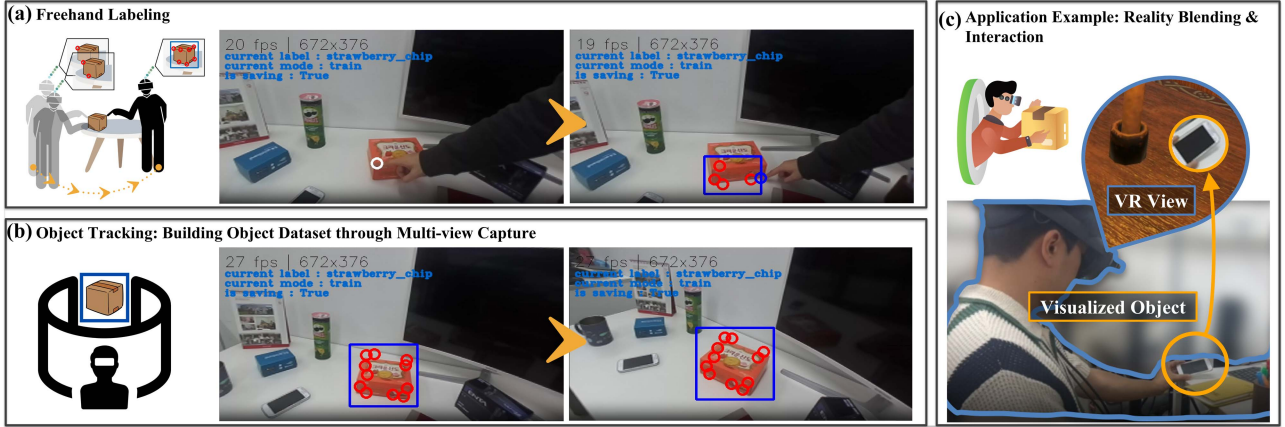


Fig. 1. Overview of the SelfBlending abstraction flow. (a) Freehand labeling where users iteratively assign points to delineate an object's boundary, generating a bounding box. (b) Object tracking and multi-view capture from different angles. (c) The user reaches for and interacts with the real-world object intuitively inside the virtual environment.

Replica-based systems [11] can support visually rich transitions through premodeled environments, but they require prescanning and rescanning when physical layouts change, limiting practicality in dynamic or personal VR settings.

Taken together, the approaches outlined above place substantial constraints on how easily users can expand and personalize the set of physical objects available for interaction during an ongoing VR experience. These constraints are especially pronounced in everyday settings, where objects, layouts, and interaction needs frequently change. Despite these research efforts to integrate physical objects into VEs, the standard mechanisms currently available to general users remain either removing the headset or using global video see-through (default passthrough). While these methods are universally accessible, they inherently compromise the user's experience: removing the headset interrupts the VR session entirely, while default passthrough substitutes the virtual world with the physical one. Therefore, to propose a solution that addresses the constraints of prior work while balancing the accessibility of standard methods with immersive continuity, we developed the SelfBlending framework. This framework provides an on-demand approach that supports users in registering and accessing physical objects with minimal overhead, rather than requiring full-scene exposure or controlled instrumentation. SelfBlending enables VR users to register surrounding physical objects via simple freehand labeling and selectively blend a specific target item into their VE using a mechanism we refer to as conditional passthrough (Fig. 1).

SelfBlending was evaluated in a within-subject user study with 18 participants during seated, single-object tabletop interactions under three conditions: (1) headset-off (HO), where participants physically removed the headset to interact with real-world objects; (2) default passthrough (DP), wherein the headset's built-in passthrough feature was used; and (3) conditional passthrough (CP), wherein SelfBlending was used. The perceived workload, sense of presence, cross-reality interaction experience, interaction time, and subjective preference of participants were measured to assess how the CP condition supports the immersive, efficient, and contextually responsive interactions of users with physical objects in the VE.

The following are the main contributions of this study: 1) a gesture-based method was proposed for freehand labeling and integration of physical objects into VEs, thereby enabling the ad-hoc enrollment of uninstrumented physical objects without relying on rigid pre-scanning or hardware-based tracking, and 2) a contextual blending interface was constructed that dynamically supported object recognition within the user's field of view, enabling intuitive physical interaction without disrupting immersion. We formulated the following questions to analyze user experience and performance across the conditions:

- *RQ1:* How does the CP condition support users in maintaining a sense of presence in the VE while accessing physical objects?
- *RQ2:* How do the interaction conditions differ in terms of perceived workload and objective performance?
- *RQ3:* How effectively do the interaction conditions support users' simultaneous awareness of both virtual and physical environments, and to what extent do these environments influence users' in-task movements, reactions, or attention during interaction?
- *RQ4:* How does the CP condition shape users' perception of interaction plausibility and preference when reflecting on their experience of accessing real-world objects during VR immersion?

II. RELATED WORK

A. Cross-Reality Interaction With Physical Objects in VR

While VR offers deeply immersive experiences, it often severs the user's connection with their immediate physical environment [12], [13]. This is particularly problematic during extended or task-intensive sessions, wherein users may need to interact with real-world objects such as typing on a keyboard or responding to bystanders [1], [14], [15]. Therefore, situational awareness must be maintained in such cases and adaptive transitions must be designed between physical and virtual worlds to preserve usability and minimize immersion disruption [1], [12]. Recent studies addressed these issues [6], [16] by selectively blending real-world elements into VR in a context-aware manner. This

enabled users to stay grounded in VR while interacting with physical objects as functional components of the virtual experience.

Cross-reality systems support this blending by integrating physical and digital environments across the virtuality continuum [4], [17], [18] but require advanced technical skills and complex setups. Recent studies have addressed this gap. For instance, VRception [19] allows developers to prototype cross-reality scenarios within VR by simulating elements from different points along the spectrum. Although this toolkit does not directly incorporate physical objects, it reflects the growing interest of researchers in accessible frameworks for mixed interactions. This shift highlights the importance of systems that enable a dynamic and bidirectional integration of physical and virtual elements.

Prior tracking methods rely on objects' proximity to the user's hand or headset [8], or use proximity information along with pose data from motion capture and RFID-equipped objects [20]. While effective for blending nearby objects into the VE, these methods limit interaction range; users must hold objects at specific distances to enable blending. They also often require pre-setup or object instrumentation. Systems based solely on proximity raise a key limit: how can users initially locate and retrieve physical objects to bring them close enough for blending? This may work for familiar items like smartphones, but broader interaction requires embedded tracking aids to support reachability and VE integration. Thus, marker-based tracking is a common method for integrating physical objects into VEs [6], [21], [22], [23], [24]. While reliable and robust in tracking, it imposes hardware burdens and is better suited for controlled setups than everyday VR contexts involving interactions with diverse physical objects. Some methods rely on image processing of surface features such as shape and color [3], [4], which often require technical knowledge, such as tuning parameters or attaching distinctive visual markers, to track or blend new objects into the VE. In this context, SelfBlending was proposed herein, which enabled freehand labeling and visual inclusion of physical objects into the VE. By eliminating the need for predefined markers, external hardware, or low-level visual matching, it empowers users to flexibly integrate and personalize real-world objects via an intuitive labeling interface and real-time artificial intelligence (AI)-based tracking.

B. Passthrough Techniques for Real-World Interaction in VR

Passthrough technology has addressed the challenge of integrating physical awareness into VEs. Studies explored how this technology enables users to remain aware of and interact with nearby physical objects in the VE [25]. Passthrough technology in VR headsets used front-facing cameras to capture a live video feed of the user's real world and displayed it; as a result, users remained aware of their surroundings without removing their headsets [26], [27]. This improvement was initially implemented as full passthrough, typically activated with a button press or a gesture [28]. It replaced the entire virtual scene with a real-world video for the users [2], [29]. However, it paused the VR application and disrupted immersion.

This evolved into passthrough-augmented virtuality, overlaying the physical world onto persistent virtual content for smoother co-engagement across spaces [1], [27].

Although passthrough videos supports user awareness and safety, it can reduce user presence, cause simulator sickness, and distort self-perception [27], [30]. More refined methods such as windowed passthrough have been proposed to address these issues, allowing users to open small, customizable views into the real world while remaining immersed in the VE [16], [31]. However, these approaches typically rely on user actions to summon customizable passthrough views and offer limited support for intuitive object-specific interaction.

To advance beyond these constraints, SelfBlending supports object-specific, context-aware inclusion of physical elements. Real-world objects can be selectively augmented into the VE on demand, allowing users to access them while maintaining immersive continuity, situational awareness, and high interaction plausibility.

III. SELFBLENDING

A. Freehand Labeling Technique

For freehand labeling, a ZED Mini camera [32] was mounted on a Meta Quest Pro [33], directly projecting its color feed onto the headset display. This allows users to perceive surroundings while labeling physical objects.

Users delineate the 2D boundary of an object of interest by pointing with their index finger (Fig. 1(a)). MediaPipe hand-tracking model [34], [35] was deployed on the ZED Mini color feed using 0.5 confidence thresholds for detection and tracking. When a user points at an object with their index finger, a white circle appears at their fingertip. Registration occurs when the user holds an extended index finger and a closed palm still for two seconds, signaled by a circle color change (Fig. 1(a)). The user can input as many points as required. After the labeling is complete, the system automatically generates a rectangular bounding box containing all the registered points (Fig. 1(a)).

B. Generating a Dataset for an Object of Interest

After initial freehand labeling (Fig. 1(a)), the system transitions into a tracking phase for capturing diverse visual representations of the labeled object. This phase enables the user to observe the object from multiple viewpoints. These views can be compiled into a dataset that accounts for variations in perspective, occlusions, and lighting conditions inherent in real-world environments (Fig. 1(b)).

The system leverages the 6-degree-of-freedom (6-DoF) data obtained from the inertial measurement unit (IMU) integrated within the ZED Mini camera for robust object tracking [32]. The IMU provides data on linear acceleration and angular velocity for estimating the camera's pose relative to the object. The tracking phase is initiated immediately after placing the first bounding box. The tracking algorithm uses a sensor fusion approach and integrates the IMU data with visual information from the camera feed using an extended Kalman filter (EKF). The EKF estimates a 12-element state vector, comprising the

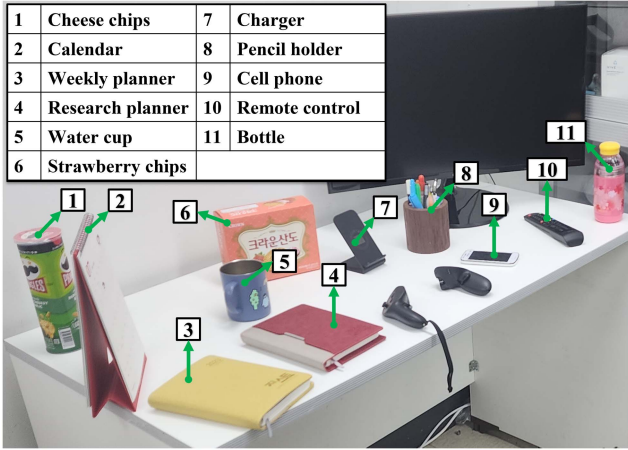


Fig. 2. Tabletop arrangement of objects used as interaction targets.

object's 3D pose and corresponding velocities, to ensure stable 2D bounding box projections. It recursively updates its estimates by predicting the state of the object based on the IMU's motion model and corrects the estimates using visual observations of the object's features. This process enhances the accuracy and stability of object tracking even in the presence of partial occlusions or dynamic lighting conditions. This labeling and tracking technique builds on our previous work, which introduced a freehand interaction method for labeling objects in both 2D and 3D coordinates, particularly in the context of human-robot interaction [36].

C. Dataset Aggregation and Model Training

The label information of the collected labeled images is then stored in corresponding annotation files. Each annotation is formatted according to the You Only Look Once (YOLO) standard [37], which encodes the object class, bounding box center coordinates, width, and height and normalizes them relative to the image dimensions. The YOLOv5 model was used because it offered superior balance between inference speed and accuracy [38], [39]. The dataset containing RGB images and YOLO-formatted annotations was used to fine-tune the YOLOv5 model at an internal 640x640 resolution for 100 epochs (batch size 16) with an 80/10/10 split for recognizing user-defined objects. This model was trained locally on the same PC used in the user study in Section IV. The pipeline, including freehand labeling (Section III-A), dataset generation (Section III-B), and model training was executed by the experimenter prior to the study. This pre-configuration ensured that all participants interacted with a model with uniform detection accuracy, isolating the effects of the interaction conditions (HO, DP, CP) from potential variances arising from individual differences in labeling and dataset collection.

D. Object Set

Various commonly used desktop items were arranged as example interaction targets, as shown in (Fig. 2). The personalization capability of SelfBlending beyond conventional class

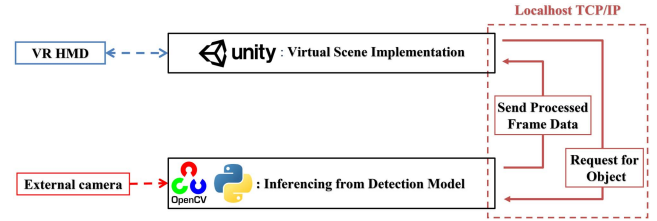


Fig. 3. Overview of the real-time detection pipeline: the Unity VR application sends a TCP request to the Python-based detection module, which processes frames from the external camera and returns the bounding box data to render CP visualization around the selected object.

ontologies that are typically used in YOLO-based models was demonstrated. To this end, custom labels were assigned to two visually similar notebooks (e.g., a weekly planner and a research planner). All objects were annotated using the aforementioned freehand labeling method, ensuring flexibility through the easy enrollment of new items. The trained models tracked the target object for blending, supporting personalization by distinguishing specific items via custom labels.

E. Real-Time Object Detection and Rendering

After model training, the system performed real-time object detection for integration into the VE. The CP condition was defined as a scenario in which the user intends to interact with a specific real-world object and only that object is selectively rendered within the VE. For this condition, a ZED Mini camera was mounted on the headset to provide an egocentric viewpoint aligned with the user's perspective; this addition was inspired by previous designs for video see-through (VST) head-mounted displays [40], [41]. This camera was also mounted because Meta Quest Pro does not permit access to raw camera streams for third-party apps, and developers only receive higher-level passthrough data (such as skeletons and meshes), not the raw RGB feed [42].

The real-time CP condition data flow and control is shown in (Fig. 3). To support the interaction between the detection module and the Unity-based VR environment, a two-way transmission control protocol (TCP)/Internet protocol (IP) communication channel was implemented over the local host. Using this interface, Unity sent object requests to the Python module that contained the incoming video feed from the external camera using the trained object detection model. The processed frames that showed detected objects with bounding boxes were then returned to Unity for visualization. The video feed from the external camera was processed at a resolution of 672×376 pixels and a refresh rate of 60 Hz, maintaining real-time performance of the system while leveraging the wide field of view of the camera. Each processed frame was passed through the trained model to extract the bounding box's center coordinates and dimensions, along with the confidence score of the detected class corresponding to the requested object. These parameters were then used to generate a masked output (a circular visualization window) that dynamically scales to circumscribe the detected bounding box, rendered with a white background (Fig. 4).

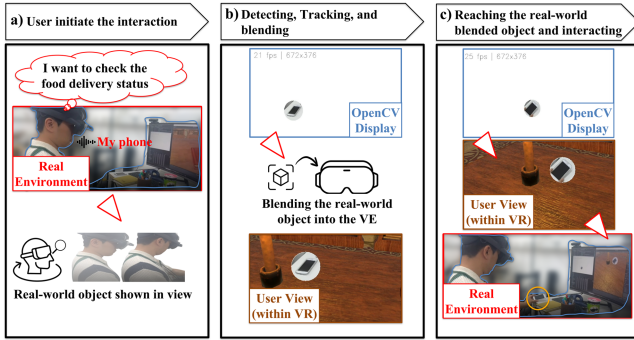


Fig. 4. Step-by-step demonstration of the CP condition. (a) The user initiates interaction by speaking the name of the target object. (b) The trained model detects and localizes the real-world object, which is then rendered in real-time within the VE. (c) The user reaches for and interacts with the real-world object while remaining immersed in the VE.

The processed frames were received via the established TCP connection and rendered onto a plane positioned directly in front of the center-eye anchor of the VR headset. The resolution ratio of the rendering plane was matched to the dimensions of the incoming frame to ensure consistency in visual mapping. The placement of the rendering plane was manually adjusted to calibrate the perceived distance of the passthrough content. A custom shader was also implemented to selectively render the white background as transparent and preserve any bright pixels within the circular passthrough region (Fig. 4). Users interacted with the detected real-world objects that were seamlessly blended into the VE via a dynamic, context-aware passthrough window; this configuration is referred to as the CP condition herein.

IV. EXPERIMENT DESIGN

This study does not evaluate the usability of SelfBlending from a system setup or configuration perspective. Rather, our focus lies in investigating the efficacy of the proposed SelfBlending framework (represented as the CP condition) in supporting users to interact with physical objects while being actively immersed in a VE. To this end, participants were exposed to three different interaction conditions that are commonly used for establishing real-object interaction in VR. (1) HO condition, wherein participants physically removed the headset; (2) DP condition, wherein the headset's built-in passthrough feature was used; and (3) CP condition, wherein SelfBlending was used.

A. Demographic Information

We recruited 18 participants (4 females and 14 males), aged between 19 and 34 years ($M = 25.72$, $SD = 3.53$). None of the participants reported any mobility issues that could limit their participation. Four participants had no prior VR experience. All participants adhered to all the experimental conditions in a within-subjects design. Each participant received a \$20 compensation for their two-hour participation. The experimental protocol was approved by our research institute (IRB No. 20250109-HR-80-06-04).

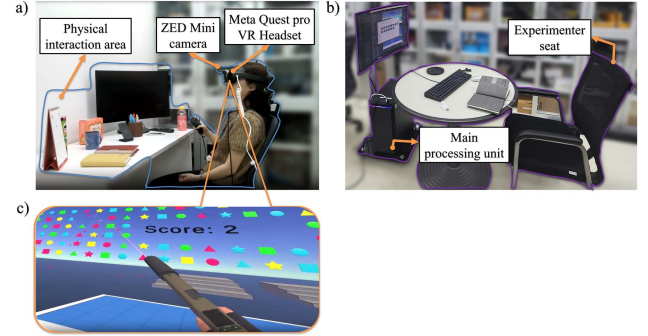


Fig. 5. Overview of the experimental setup. (a) User wearing a Meta Quest Pro with a front-mounted ZED Mini camera. (b) Tabletop arrangement of items. (c) Shooting game environment.

B. Experiment Setup

Participants were seated in a swivel chair while wearing a Meta Quest Pro VR headset equipped with a front-mounted ZED Mini RGBD camera (Fig. 5(a)). Both devices were wired and connected to a desktop PC placed behind the participants' chairs. The PC featured an Intel(R) Core(TM) i9-10900 K CPU @ 3.70 GHz and an NVIDIA GeForce RTX 3090 GPU, configured in a CUDA-enabled environment for real-time RGBD camera processing. Physical objects were arranged on a table placed in front of participants, and the system was pretrained using SelfBlending to recognize and track these items. An additional desk was positioned behind the participants (Fig. 5(b)), where the experimenter monitored the system performance and controlled the start and end times of each session.

A visually engaging shooting game designed to maintain user immersion and sustained focus was used as the virtual content (Fig. 5(c)). This continuous task flow simulated an uninterrupted VR experience, in which accessing real-world objects was critical to the objective of the study: examining how users interact with physical items while remaining immersed. This design choice aligned with recent studies on cross-reality interaction, where game-based environments supported more natural transitions between realities and sustained user engagement even under varying conditions [43], [44]. Feld et al. [45] used a gamified memory task to evaluate cross-reality transitions and found that such immersive tasks helped participants to remain engaged and focused. They thus preserved the ecological validity of transitions between virtual and physical contexts.

C. Procedure

Participants were seated in front of a desk on which the predefined physical objects were arranged. Each participant first completed a demographic and consent form before being introduced to the experimental setup. The experimenter explained the equipment and task flow, guiding the participants to use the Meta Quest Pro headset, VR controller, and interaction mechanisms (Fig. 5(a)).

Participants then engaged in a training session designed to familiarize them with the three conditions and the VR shooting game (Fig. 5). During training, they practiced using the DP

condition by double-tapping the side of the headset to enter and exit the passthrough mode. They also learned to activate the CP condition by vocally calling out the name of a physical object and to terminate the passthrough mode using the voice command “stop.” The voice activation feature was implemented using a Wizard-of-Oz approach, wherein the experimenter manually triggered the appearance of the selected object by checking a box on the Unity interface and deactivated it upon hearing “stop.” This manual control did not affect the autonomy of SelfBlending; it continued to detect, track, and blend the physical object into the VE. Participants were also trained to use the VR controller to shoot colored geometric targets within the game (Fig. 5(a)).

They were informed that each gameplay session would continue until they had shot all of the virtual objects. To simulate scenarios in which participants may spontaneously need to access nearby physical items while being immersed in VR, they were instructed to initiate a real-world interaction at a self-determined moment, typically after engaging with what they estimated to be around half of the colored geometric targets. This approximate mid-task point was used to ensure that participants were familiarized with the environment and were sufficiently immersed in the shooting game before their experience was interrupted to interact with a physical object. After the interaction was triggered, the VR session automatically paused when participants removed their headset in the HO condition, double-tapped the HMD in the DP condition, or vocalized the object name in the CP condition. After ending the interaction with the real object, the game resumed when the participants wore their headsets in the HO condition, double-tapped the HMD in the DP condition, or called out “stop” in the CP condition. They then continued to play the game for the remainder of the session. The interaction time was calculated as the duration between these pause and resume events.

The duration of the session varied across participants based on their shooting speed and accuracy as well as the time required to establish a physical interaction under each condition. To reduce the number of confounders, each participant selected a single physical object in the first condition and interacted with the same object in all subsequent conditions. This approach helped to minimize the variability in the location, familiarity, and accessibility of an object, which could affect measures such as the interaction time and subjective perception of participants. This control improved the internal validity by keeping the real-world task consistent across all conditions; however, it introduced a degree of artificiality compared with the naturalistic VR use, wherein users interact with a range of objects as required. However, this trade-off supported a focused, within-subject comparison aimed at isolating the influence of interaction techniques rather than object-specific factors.

All participants completed the experiment under the HO, DP, and CP conditions. The condition exposure was counterbalanced across participants to mitigate order effects. The interaction task and VR environment remained constant across all sessions, and only the method used to access and interact with the physical object varied. After the experiment, participants filled out questionnaires assessing their perceived workload, presence, and

cross-reality interaction experience. They provided an overall preference rating for the interaction conditions and answered open-ended questions.

D. Measures

A combination of quantitative and qualitative measures was employed to evaluate the user experience under each interaction condition. They completed an online questionnaire assessing their perceived workload, presence, and cross-reality interaction experience.

The participant workload was assessed using the NASA Task Load Index (NASA-TLX) [46] that comprises six subscales designed to derive an overall workload score: mental demand, physical demand, temporal demand, performance satisfaction, effort, and frustration. User presence was measured using the Igroup presence questionnaire (IPQ) [47] that contains 14 items spanning four subscales: general presence, spatial presence, involvement, and realism.

Subscales were also selected from the cross-reality interaction questionnaire (CRIQ) [2], which is a purpose-built questionnaire developed to assess how users perceive interactions that span across physical and VEs. The CRIQ captures key experiential aspects of cross-reality transitions, wherein users mentally synchronize elements from both realities during an ongoing immersive task. It is a well-aligned tool for assessing the effectiveness of SelfBlending in supporting the seamless integration between physical environment and VE when users interact with real-world physical objects. All items are rated on a 7-point Likert scale. Herein, three CRIQ subscales, namely awareness, behavioral influence, and plausibility illusion, were considered. The CRIQ/awareness subscale measured users’ perceptual access to the physical environment and VE during an interaction. The CRIQ/behavioral influence subscale captured the degree of influence of each environment on the user behavior during an interaction. The CRIQ/plausibility illusion subscale evaluated the perceived coherence, synchronicity, and responsiveness of the interaction experience.

Participants rated their overall preference for the three interaction conditions using a 5-point Likert scale, indicating how each condition supported their interaction with real-world objects while maintaining immersion in the VE. They were also invited to provide open-ended feedback via written comments or verbal discussion with the experimenter to share observations and suggestions for improvement.

To ensure consistent measurements, the CRIQ/general subscale was excluded because it overlapped with the general preference ratings. This subscale assesses broad usability aspects such as disruption, efficiency, and comfort. The CRIQ/place illusion subscale, which focuses on shifts in user presence before, during, and after the interaction, was also excluded because presence was already comprehensively measured using the standardized IPQ.

In addition to these subjective assessments, the time taken by participants to reach the selected physical object and interact with it was recorded. This time was calculated as the duration between the VR game’s pause and resume events, triggered

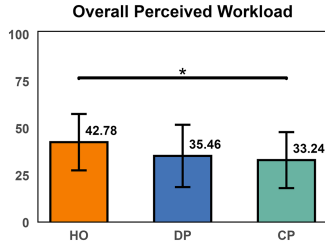


Fig. 6. Perceived workload (NASA-TLX) across the three conditions; significance level are indicated as $*p < 0.05$.

by predefined participant actions specified for each condition (Section IV-C).

V. RESULTS

The quantitative and qualitative results derived from participants' responses and behavior across HO, DP, and CP conditions were analyzed; these included results obtained from the standardized questionnaires (NASA-TLX, IPQ, and selected CRIQ subscales), overall condition preference ratings, open-ended participant feedback, and objective measurements of interaction duration. All statistical tests were conducted using one-way repeated measures (RM) ANOVA with Bonferroni post-hoc comparisons. Regarding the analysis of our Likert-type responses, we verified the assumptions of normality and sphericity using the Shapiro-Wilk and Mauchly's tests, respectively. We adhered to established guidance [48], [49] demonstrating that parametric tests, such as one-way RM ANOVA, are robust for analyzing items with five or more response categories, even when the underlying data is ordinal. With $N = 18$ ($\alpha = .05$, $1 - \beta = .80$), sensitivity analysis confirmed detection of medium-to-large effects (Cohen's $f \approx 0.24 - 0.40$), sufficient for the observed differences across our metrics.

A. User-Reported Experience Across Different Conditions

1) *Perceived Workload*: The participants' overall perceived workload was measured using the NASA-TLX questionnaire, and the weighted scores were averaged across its six subscales. As shown in (Fig. 6), the HO condition was associated with the highest workload ($M = 42.78$, $SD = 14.99$), followed by the DP condition ($M = 35.46$, $SD = 16.57$); the CP condition showed the lowest workload ($M = 33.24$, $SD = 14.89$). The one-way RM ANOVA revealed a significant impact of condition on workload, $F(2, 34) = 3.38$, $p = .046$, $\eta^2 = .166$. Post hoc comparisons with Bonferroni correction indicated that the CP condition elicited a significantly lower workload than the HO condition ($p = .024$, mean difference = 9.54). No significant differences in the workload were observed between the HO and DP conditions ($p = .199$) or between DP and CP conditions ($p = 1.000$). These results indicate that the CP condition was perceived as less demanding than the HO condition by the participants when interacting with physical objects. NASA-TLX subscale analysis revealed that this lower perceived demand was driven by significantly lower frustration ($F(2, 34) = 4.39$, $p = .02$, $\eta^2 = .21$) and temporal demand

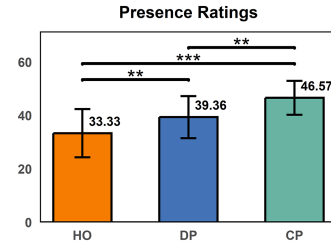


Fig. 7. Presence rating across the three conditions; significance levels are indicated as $**p < 0.01$.

($F(2, 34) = 4.67$, $p = .016$, $\eta^2 = .22$) in CP compared to HO (post-hoc: $p = .016$ and $p = .048$, respectively), with no significant differences in other subscales.

2) *Presence*: Presence, defined as the sense of 'being there' in the VE [50], was evaluated using the IPQ to measure how well each condition maintained engagement during real-world object interactions.

The one-way RM ANOVA revealed a significant effect of conditions on presence, $F(2, 34) = 19.876$, $p < .001$, $\eta^2 = .539$, indicating a large effect size. As shown in (Fig. 7), participants reported the lowest sense of presence in the HO condition ($M = 33.33$, $SD = 9.02$). Presence scores increased in the DP condition ($M = 39.56$, $SD = 7.87$) and were the highest in the CP condition ($M = 46.59$, $SD = 6.32$). Post hoc comparisons revealed that the CP condition significantly outperformed both the HO ($p < .001$) and DP ($p = .006$) conditions. The DP condition also led to a significantly higher presence than the HO condition ($p = .010$). IPQ subscale analysis [51] revealed that the CP condition elicited significantly higher spatial presence ($F(2, 34) = 12.10$, $p < .001$, $\eta^2 = .42$), realism ($F(2, 34) = 13.42$, $p < .001$, $\eta^2 = .44$), and involvement ($F(2, 34) = 8.90$, $p < .001$, $\eta^2 = .34$) compared to both HO (post-hoc: $p < .001$, $p < .001$, and $p = .008$, respectively) and DP (post-hoc: $p = .030$, $p = .022$, and $p = .039$, respectively). The DP condition also showed significantly higher realism ($p = .018$) and involvement ($p = .045$) than HO, but no significant difference was observed for Spatial Presence ($p = .110$). General presence ($F(2, 34) = 5.42$, $p = .009$, $\eta^2 = .24$) was significantly higher in CP compared to HO ($p = .007$), with no significant differences found between CP and DP ($p = .214$) or DP and HO ($p = .492$).

These findings suggest that the CP condition most effectively sustained the sense of presence, preserving spatial presence, realism, and involvement. The manual interruption in HO and the visual switch in DP disrupted this continuity, whereas CP allowed physical interaction while remaining psychologically anchored within the VE.

3) *Cross-Reality Experience: Virtual and Physical Awareness* Virtual awareness, defined as the user's perceptual access to the VE during a physical interaction, was evaluated using the CRIQ question "During the interaction, how aware were you of the VE/entities?" This measure assessed the extent to which each condition allowed participants to remain mentally situated in the VE while accessing the real-world content.

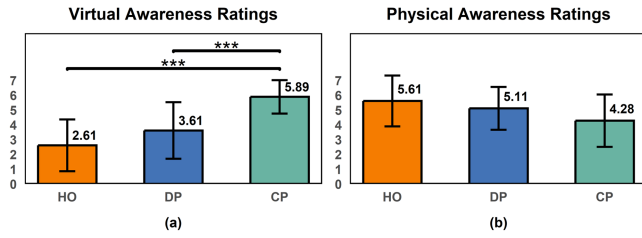


Fig. 8. Virtual and physical awareness ratings across the three conditions. (a) Virtual awareness rating. (b) Physical awareness ratings; significance level is indicated as *** $p < 0.001$.

The one-way RM ANOVA revealed a significant effect of condition on virtual awareness, $F(2, 34) = 27.382$, $p < .001$, $\eta^2 = .617$. As shown in (Fig. 8(a)), participants reported the lowest virtual awareness in the HO condition ($M = 2.61$, $SD = 1.75$), followed by the DP condition ($M = 3.61$, $SD = 1.91$); the highest score was reported for the CP condition ($M = 5.89$, $SD = 1.13$). Post hoc comparisons showed significant increases from HO to DP ($p = .026$), from HO to CP ($p < .001$), and from DP to CP conditions ($p < .001$). This indicated that the CP condition significantly enhanced users' ability to stay aware of the VE compared to the DP condition. These findings indicate that the CP condition was the most effective in maintaining virtual awareness during real-world interaction, compared to the HO and DP conditions.

Physical awareness, defined as the user's perceptual access to the physical environment during an interaction, was evaluated using the CRIQ question "During the interaction, how aware were you of the physical environment/entities?" This measure assessed how effectively each condition enabled users to remain connected to their real-world surroundings while interacting with the VE.

The one-way RM ANOVA revealed no significant effect of condition on physical awareness, $F(2, 34) = 3.192$, $p = .054$, $\eta^2 = .158$. As shown in (Fig. 8(b)), physical awareness was the highest in the HO condition ($M = 5.61$, $SD = 1.72$), slightly lower in the DP condition ($M = 5.11$, $SD = 1.45$), and the lowest in the CP condition ($M = 4.28$, $SD = 1.78$). The overall test approached significance; however, post hoc comparisons did not reveal significant pairwise differences ($p > .05$). Thus, although the CP condition provided a limited object-focused passthrough window, it maintained a comparable level of physical awareness relative to the DP and HO conditions. The overall difference across conditions was not statistically significant; however, the CP condition offered moderate physical awareness without requiring full visibility of the real world. This reinforced the effectiveness of SelfBlending in enabling targeted, context-aware physical interaction while preserving high virtual situational awareness.

Virtual and Physical Behavioral Influence: Virtual behavioral influence, defined as the degree to which a user's behavior is influenced by the VE during a physical interaction, was evaluated using the CRIQ question "During the interaction, how much was your behavior influenced by the VE/entities?"

The one-way RM ANOVA revealed a significant effect of condition on virtual behavioral influence, $F(2, 34) = 14.426$,

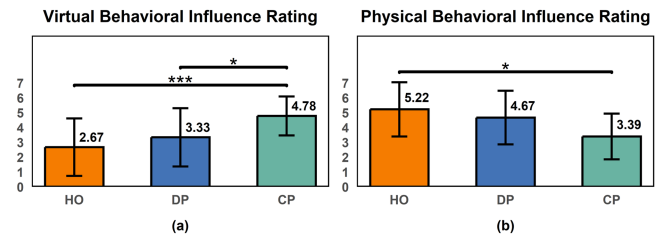


Fig. 9. Virtual and physical behavioral influence ratings across the three conditions. (a) Virtual behavioral influence rating and (b) Physical behavioral influence rating; significance levels are indicated as * $p < 0.05$ and *** $p < 0.001$.

$p < .001$, $\eta^2 = .459$. As shown in (Fig. 9(a)), participants reported the lowest influence in the HO condition ($M = 2.67$, $SD = 1.94$), followed by the DP condition ($M = 3.33$, $SD = 1.97$); the highest influence was observed in the CP condition ($M = 4.78$, $SD = 1.31$). Post hoc tests confirmed that the CP condition led to significantly higher behavioral influence than both the DP ($p = .004$) and HO ($p < .001$) conditions, whereas the difference between HO and DP conditions was not significant ($p = .185$). These findings demonstrate that the CP condition most effectively preserved users' behavioral engagement with the VE during a physical interaction; it outperformed the DP and HO conditions in maintaining user actions shaped by the virtual context.

Physical behavioral influence, defined as the degree to which the user's behavior was influenced by the physical environment during interaction, was evaluated using the CRIQ question "During the interaction, how much was your behavior influenced by the physical environment/entities?"

The one-way RM ANOVA revealed a significant effect of condition on physical behavioral influence, $F(2, 34) = 5.825$, $p = .007$, $\eta^2 = .255$. As shown in (Fig. 9(b)), participants reported the highest physical behavioral influence in the HO condition ($M = 5.22$, $SD = 1.83$), followed by the DP condition ($M = 4.67$, $SD = 1.82$), and the lowest in the CP condition ($M = 3.39$, $SD = 1.54$). Post hoc comparisons indicated that physical behavioral influence in the CP condition was significantly lower than that in the HO condition ($p = .019$). The differences between HO and DP ($p = .789$) and between DP and CP conditions ($p = .122$) were not statistically significant.

The CP condition provided only a narrow, object-focused passthrough; however, it offered a level of physical behavioral influence that was statistically comparable to the fully transparent DP condition ($p = .122$). Notably, the DP condition also failed to match the HO condition in terms of physical influence, suggesting that full passthrough alone did not replicate the behavioral shift caused by removing the headset. This indicated that the CP condition enabled an effective interaction of users with physical objects without overwhelming their virtual experience, offering a balanced form of cross-reality engagement.

Plausibility Illusion: Plausibility illusion, defined as the user's overall perception of the cross-reality interaction as coherent, synchronized, and responsive, was evaluated by averaging responses to three CRIQ questions: "How much did the interaction feel plausible to you?," "How much did you have the perception

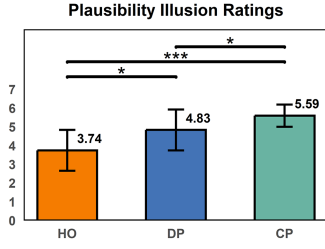


Fig. 10. Plausibility illusion ratings across the three conditions; significance levels are indicated as $*p < 0.05$ and $***p < 0.001$.

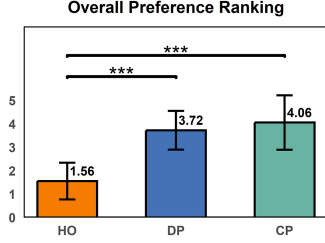


Fig. 11. Overall preference ratings across the three conditions; significance level is indicated as $***p < 0.001$.

of a synchronized interaction between the physical and VE?,” and “How much did you feel like the experience was responding to you (interactive)?”

The one-way RM ANOVA revealed a significant effect of condition on plausibility illusion, $F(2, 34) = 16.735, p < .001, \eta^2 = .496$. As shown in (Fig. 10), participants reported the lowest plausibility in the HO condition ($M = 3.74, SD = 1.09$), higher in the DP condition ($M = 4.83, SD = 1.09$), and the highest in the CP condition ($M = 5.59, SD = 0.59$). Post hoc comparisons showed significant differences between HO and DP ($p = .021$), HO and CP ($p < .001$), and DP and CP conditions ($p = .042$). These results indicated that the CP condition most effectively supported a plausible cross-reality experience and outperformed both the HO and DP conditions in perceived realism, synchronicity, and responsiveness.

4) *Overall Preference*: Participants rated their overall preference for each condition on a 5-point Likert scale (1 = very low, 5 = very high) after completing experiments under the HO, DP, and CP conditions. This rating reflected their suitability and comfort in using the interaction conditions for accessing real-world objects during VR use.

The one-way RM ANOVA revealed a significant difference in overall preference ratings, $F(2, 34) = 29.417, p < .001, \eta^2 = .634$, as shown in (Fig. 11). The CP condition received the highest preference rating ($M = 4.056, SD = 1.162$), followed by DP ($M = 3.722, SD = 0.826$) and HO conditions ($M = 1.556, SD = 0.734$). Post hoc comparisons showed that both CP and DP conditions were rated significantly higher than the HO conditions (both $p < .001$), whereas no significant difference was found between the CP and DP conditions ($p = 1.000$). These results suggested a clear participant preference for passthrough-based interaction methods over the HO approach when accessing real-world objects during VR experiences.

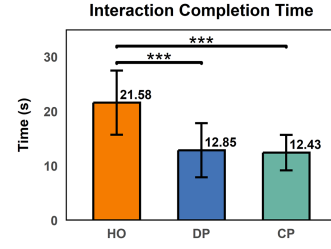


Fig. 12. Interaction completion time across the three conditions; significance level is indicated as $***p < 0.001$.

B. Quantifying Interaction Impact

The interaction time of each participant across the HO, CP, and DP conditions was measured to examine which interaction mechanism enabled the quickest access to real-world objects. Interaction time is defined as the duration between the initiation and termination of each interaction episode (Section IV-D). This included the moment the participant triggered the interaction to the moment they completed the real-world task and exited the interaction mode.

The one-way RM ANOVA revealed a significant effect of condition on the overall interaction time, $F(2, 34) = 37.35, p < .001, \eta^2 = .687$. As shown in Fig. 12, the mean interaction time in the HO condition ($M = 21.58, SD = 5.81$) was significantly longer than those in the DP condition ($M = 12.85, SD = 5.00, p < .001$) and CP condition ($M = 12.43, SD = 3.25, p < .001$). No significant difference was found between the DP and CP conditions ($p = .087$), indicating that both passthrough-based methods enabled faster interaction with real-world objects compared to the HO condition.

VI. DISCUSSION

A. Addressing Research Questions

1) *RQ1: How does the CP condition support users in maintaining a sense of presence in the VE while accessing physical objects?*: Within the scope of our seated tabletop setup, the CP condition most effectively preserved users’ sense of presence and immersion by reducing perceptual disruption during physical interaction, outperforming both the HO and DP conditions. This conclusion was supported by the subjective presence scores, where the CP condition significantly outperformed both the HO and DP conditions (Fig. 7), yielding a large effect size ($\eta^2 = .539$).

Thus, users experienced psychological continuity under the CP condition. This is corroborated by the significantly higher spatial presence in CP compared to DP, which confirms that users felt situated in the VE even while reaching for real objects. Psychological continuity is a cognitive state in which users remain steadily engaged with the VE without any perceptual or attentional disruption; this concept is grounded in Slater’s definition of presence as emerging from uninterrupted sensorimotor loops between perception and action [52]. When these loops are broken, such as by removing the headset in the HO condition, users are forced to shift cognitive frames that can reduce their immersion.

The DP condition was less disruptive than the HO condition; however, it disrupted psychological continuity as well as users' flow of engagement while weakening their sense of presence. The lower involvement and realism scores in the DP condition indicate that the abrupt visual switch acted as a distraction rather than a supportive interaction aid. These observations were similar to previous studies [45], [53], wherein full passthrough views, although visually rich, failed to anchor the user in the virtual context due to unfiltered, incongruent real-world content. Guo et al. [27] confirmed that while passthrough views boosted real-world awareness, they offered poor immersion.

In contrast, object-focused blending in the CP condition limited users' exposure to only task-relevant real-world information, preserving the integrity of the virtual scene. This targeted approach resonated with the findings reported in a previous study [45], wherein selective, coherent transitions improved user presence in cross-reality tasks. Other similar studies [54] and [16] found that context-aware and customizable overlays, rather than full passthrough, yielded enhanced immersion by reducing attentional switching and preserving the user's virtual context.

2) *RQ2: How do the interaction conditions differ in terms of perceived workload and objective performance?*: The lowest overall workload was observed in the CP condition, significantly lower than the HO condition (driven by significant reductions in frustration and temporal demand) and comparable to the DP condition (Fig. 6), yielding an effect size of ($\eta^2 = .166$).

This reduction in subjective temporal demand aligns with the objective interaction time results, showing CP enabled seated tabletop single-object interaction as rapidly as DP and faster than HO (Fig. 12). By eliminating the friction of manual transitions required in HO, the CP condition reduced both the perceived time pressure and the frustration associated with such disruptive physical transitions.

Although the CP condition offered less persistent visual access, it was comparable to the DP condition in terms of interaction speed and overall workload. This indicated that SelfBlending's on-demand, context-aware blending strategy can achieve high interaction efficiency without requiring full visual exposure to the physical environment.

Feld et al. [55] also reported similar results and showed that users preferred localized, low-disruption transitions such as portals over continuous or abrupt alternatives because they reduced the workload and enabled faster task performance across cross-reality tasks.

3) *RQ3: How effectively do the interaction conditions support users' simultaneous awareness of both virtual and physical environments, and to what extent do these environments influence users' in-task movements, reactions, or attention during interaction?*: The CP condition balanced situational awareness and behavioral continuity during seated tabletop single-object interaction, helping users remain perceptually grounded in both the virtual and physical environments without requiring a full passthrough view or exiting the VE.

A compelling double dissociation was observed between the two awareness components in cross-reality awareness. As shown in (Fig. 8(a)), the CP condition considerably enhanced virtual awareness and enabled users to maintain strong perceptual continuity with the VE even while engaging with physical objects. In contrast, the HO condition severely disrupted this connection by removing users entirely from the VE, whereas the DP condition diluted users' awareness in the virtual context although it was visually transparent. Thus, SelfBlending preserved virtual situational awareness, which a core requirement for achieving immersive continuity in cross-reality systems.

Physical awareness (Fig. 8(b)) did not differ considerably across conditions. Notably, the CP condition maintained a moderate level of physical situational awareness although only revealing a selective, object-focused view of the physical world. This highlighted the effectiveness of the context-aware blending of the CP condition. It delivers sufficient real-world cues to support user interaction without fully exposing the physical environment or breaking immersion. The HO and DP conditions presented broader real-world visibility: the former offered complete exposure and the latter offered a continuous front-facing passthrough. However, neither of these conditions yielded considerably higher physical awareness compared with the CP condition. This indicated that SelfBlending could achieve perceptual grounding and real-world access with minimal visual intrusion, contrasting with the notion that higher visual fidelity or uninterrupted passthrough is a prerequisite for achieving effective physical awareness. This idea is often implied in systems such as Passthrough+ [56] and NeuralPassthrough [57] that emphasize visual realism over interaction context.

The results for behavioral influence revealed that the CP condition was the most effective in shaping user actions around the virtual task context and minimized unnecessary shifts toward the physical world. As shown in (Fig. 9(a)), the CP condition considerably increased virtual behavioral influence compared with both the HO and DP conditions. This indicated that user behaviors were more guided by the VE even during physical interaction and that the CP condition enabled a seamless integration of physical and virtual cues. It thus supported task-directed interaction without cognitive detachment of users from the virtual experience.

The results for physical behavioral influence (Fig. 9(b)) highlighted the balancing strength of the CP condition. Despite revealing only narrow, task-specific content from the real world, it achieved a level of physical influence that was statistically comparable to that achieved in the DP condition that provided full passthrough. This indicated that users could interact naturally with real-world elements under the CP condition without being behaviorally pulled away from the VE. The HO condition had the highest physical behavioral influence but failed to preserve virtual continuity. Thus, the CP condition presented a more integrated and balanced form of cross-reality interaction, enabling natural physical engagement of users without sacrificing the behavioral fidelity of the virtual task.

These findings collectively highlight the ability of SelfBlending to preserve behavioral immersion as well as virtual and physical perceptual continuity, which are essential for effective cross-reality systems. It offers a localized, task-aware, and context-sensitive blending strategy that is aligned with users' actions within VEs. Systems such as RealityCheck [58] allow users to blend portions of their physical surroundings into VR to maintain presence and ensure safety; however, they raise an important open question: in such user-tunable systems, is behavioral and situational awareness more influenced by the virtual or the physical world? How should this balance be tuned? RealityCheck prioritizes situational responsiveness and user-configurable passthrough blending. In contrast, SelfBlending is not safety-driven but task-centered and is designed for routine VR use in stable and uncluttered environments, wherein users seek immersive continuity and occasionally require precise interaction with real-world objects. By revealing only task-relevant physical elements in real time, SelfBlending shifts the design goal toward favoring virtual continuity while enabling physical awareness when functionally required. It thus achieves a comparable sense of spatial orientation without the visual overload of full passthrough. Nevertheless, we acknowledge a trade-off regarding physical awareness and behavioral influence. While prioritizing stationary, object-specific interaction, the CP condition limits broader environmental monitoring. This isolation presents a functional limitation for contexts demanding rich real-world interaction. Thus, applications requiring broader environmental exposure or dynamic, unstructured interaction may necessitate the comprehensive connectivity of the HO and DP conditions.

4) *RQ4: How does the cp condition shape users' perception of interaction plausibility and preference when reflecting on their experience of accessing real-world objects during VR immersion?*: The results for plausibility illusion and preference ratings revealed the influence of each condition on the users' preferred method for temporarily interacting with physical objects while immersed in VR. Participants reported significantly higher interaction plausibility in the CP condition, followed by the DP condition, and the lowest in the HO condition, (Fig. 10). This indicated that users found SelfBlending to be the most coherent, synchronized, and responsive. Subjective preference ratings significantly favored the CP condition similar to that observed in plausibility scores (Fig. 11).

To better understand this pattern of preference, we examined participants' responses to an open-ended question on explaining their ratings. P4 explained, "The method using AI is the most preferred because it does not break one's concentration." Others echoed this sentiment, highlighting how the CP condition preserved users' immersion and minimized effort. For instance, P5 stated that "This AI method helped me to maintain immersion better," and P6 noted that "It was easier and smoother to use compared to the others." Similarly, P8 remarked that "I felt the least distracted when interacting with the object through a small window inside the VE," and P9 said that "It kept me more immersed and was the most convenient." In terms of efficiency, both P2 and P17 praised the CP condition for streamlining the interaction process. P2 wrote, "Saved time and effort in

addition to the smoothness of the whole process," whereas P17 commented, "It felt efficient, and I liked not having to leave the VR experience."

In contrast, the DP condition was viewed as more familiar, particularly by experienced VR users. P14 noted that "it was easy to change between real and virtual," whereas P15 reported that "It worked as expected for someone used to passthrough." However, there were indications of disruption in concentration: P10 stated that "distraction happens when tapping the HMD," and P18 mentioned "I lost my focus when the screen changed." Technical issues were also raised, with P13 commenting that "the screen resolution and field of view were limited."

The HO condition was consistently described as the most disruptive and effortful interaction method. Participants (P1,P2,P4,P8,P10,P16,P18) found it both physically and mentally taxing. P1 stated that "A bit annoying to remove the headset each time," and P10 described it as "Super distracting physically and mentally." Many participants emphasized how this method disrupted their sense of immersion: P4 said that "It broke my immersion completely." P8 commented that "This was the most inconvenient method," P9 noted that "Taking the headset off interrupted everything," and P13 added "Awareness of surroundings was better, but at the cost of immersion." Although a few participants (P12,P13), acknowledged that the HO condition provided complete awareness of the real-world environment, they agreed that this benefit came with a notable trade-off: disconnection from the virtual experience.

B. Extending the Design Space of Cross-Reality Interaction With SelfBlending

SelfBlending is a cross-reality interaction framework for scenarios wherein users intentionally engage with the physical world during an ongoing VR experience. RealityLens [16] allowed users to overlay selected portions of the real world into the VE using custom-shaped lenses; however, its workflow was controller-dependent and required users to manually activate passthrough, define lens regions, and manipulate them in the 3D space using VR controllers. This process was time-consuming and unintuitive for non-expert users. In addition, aligning moving objects or ensuring accurate blending in dynamic settings often required external tracking hardware or visual markers, which limited the accessibility and scalability of RealityLens. Although this method offers flexibility, it is fundamentally built for explicit, user-curated framing of reality, rather than seamless, embodied interaction. In contrast, SelfBlending allows users to pretrain object detection models for frequently used physical objects via a one-time, freehand image collection step. Once the model is trained, these objects can be contextually added into the VR experience by simply calling their names; this process enabled intentional and intuitive blending without using controllers and markers or via manual spatial manipulation. This design is particularly suited for personal VR environments wherein users repeatedly interact with specific physical objects (such as a bottle, keyboard, or tool). Rather than managing visual lenses or framing regions, SelfBlending offers task-relevant,

on-demand object integration that preserves users' immersion and reduces interaction overhead.

Furthermore, systems like "GradualReality" [6] introduces interaction-aware visual transitions for prescripted objects but relies on manually preconfigured proxies, boundaries, and task states. SelfBlending advances this procedure by enabling user-initiated blending of physical objects on demand without any predefined markers or zones. On the other hand, Replica-enhanced transitions in "From Real to Virtual" [11] rely on building immersive scene bridges using premodeled replicas of physical environments. Although this approach is visually rich, it requires prior 3D modeling as well as group-based object scheduling and is limited to environments that can be fully scanned. SelfBlending avoids rigid scene-level dependencies by enabling per-object blending based on real-time recognition; it is thus ideal for personal VR environments wherein users repeatedly interact with familiar objects.

Unlike previously proposed approaches, SelfBlending supports spontaneous, localized interaction with real-world elements without requiring infrastructure overhead, rigid transitions, or environmental reconstruction. These design choices expand the cross-reality interaction space by prioritizing fluidity, personalization, and minimal setup, enabling broader integration of SelfBlending into everyday VR scenarios.

C. Limitations and Future Work

While SelfBlending demonstrated the feasibility of on-demand physical object blending and interaction, achieving a seamless perceptual blend remained a challenge. P15 described the blend view as a "bubble in the virtual world," whereas P18 noted the lack of a true 3D view. These concerns aligned with prior work highlighting the difficulty of achieving perceptual blending [59] and bridging the gap between representational and perceptual realism [58]. Future works can adopt portal-style rendering and depth-aware segmentation to improve spatial clarity and immersion. Furthermore, to maintain experimental control, we employed a Wizard-of-Oz voice activation and a seated, tabletop setup. While this isolated the effects of the interaction conditions, it limits generalizability regarding automated speech recognition (ASR) latency and errors, as well as dynamic factors such as moving, occlusions, and accessing objects that are not neatly positioned. Additionally, floating passthrough windows may impose attentional costs during high-focus VR tasks (e.g., design, surgery simulations). Future work must integrate functional ASR and spatial occlusion handling to address these real-world challenges, while also determining acceptable distraction thresholds in cognitively demanding scenarios.

While the primary focus of this study was to introduce SelfBlending as a framework that enables the intuitive integration of real-world objects into the virtual environment (VE), the evaluation focused on scenarios involving a single real-world object per trial, rather than exploring multi-object blending. Although multi-object blending was not tested in this study, SelfBlending is technically capable of supporting it, as the underlying YOLO-based tracking model can detect and track all trained

items simultaneously in real time. As future work, investigating multi-object blending could provide valuable insights into how users perform complex real-world tasks while remaining engaged within the VE using SelfBlending. Furthermore, future exploration can examine how users switch between these objects naturally. Building on this idea, by installing a 360° camera, users could summon objects from any direction and allow the virtual world to help guide them toward real-world items in their surroundings. Moreover, future evaluations will extend to untethered mobile VR setups and collaborative scenarios, exploring how multiple users can share physical and virtual contexts.

In its current desktop-based implementation, SelfBlending operates at approximately 15–35 FPS during active object blending; while this is sufficient for short, object-specific interaction episodes, future work will focus on optimizing the detection pipeline and model deployment to support higher frame rates and untethered mobile VR scenarios. We also aim to unify freehand labeling, training, and object blending into a single, accessible GUI that guides non-expert users through a simple, step-by-step setup, enabling future work on how everyday users can intuitively blend personalized physical objects into VR using hand gestures and basic GUI interactions through SelfBlending.

VII. CONCLUSION

Herein, a cross-reality interaction framework known as SelfBlending that enables users to access physical objects during VR immersion via real-time, task-aware blending was proposed. Results from our seated tabletop evaluation demonstrate that SelfBlending supported immersive continuity, reduced cognitive disruption, and facilitated efficient interaction via selective, context-relevant blending. By enabling freehand gesture-based, markerless, and object-specific integration, SelfBlending offered a lightweight and flexible solution that broadened the design space for personalized and fluid cross-reality experiences.

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